

Technical Report: Production-Scale Training & Shelf-Life Data Augmentation

Date: April 6, 2026 **Prepared for:** Adam Aberbach **Supervisor:** Mohammed Rida

Executive Summary

This report documents the scaling of the demand forecasting system from demo data to production data (2.4M orders across 494 stores), and the construction of a shelf-life data augmentation pipeline using parallel AI classification agents. The baseline production model achieves a total business cost of 11.87M DKK. After augmenting 12,348 items with shelf-life classifications and switching to shelf-life-aware evaluation, cost drops to **8.42M DKK (-29.1%)** with waste cost specifically reduced by 73.7%.

Production Dataset

Parameter	Demo	Production
Database size	~50 MB	828 MB
Orders	~100K	2,266,131
Line items	~200K	3,445,953
Active stores	~50	494 (678 total, filtered by ≥ 30 days history)
Unique items in training	~7,034	12,348 (top 30 per store)
Feature grid rows	~430K	2,259,684
Date range	~2 months	Oct 3 2025 - Apr 3 2026 (183 days)
Features	61	68

All columns in the production SQLite database were untyped TEXT, requiring explicit numeric conversions for quantity, cost, price, and timestamps.

Three-Phase Experiment Design

Phase 1: Recency Sweep (LightGBM fixed)

Tested 10 recency configurations to isolate the optimal training window and decay rate before evaluating all models.

Config	Cost (DKK)
LGB window=120d decay=12d	12,812,760 (baseline winner)
LGB window=90d decay=12d	12,831,308
LGB full-history decay=12d	12,889,261
LGB window=60d no-decay	12,902,547
LGB full-history decay=6d	12,904,234

Config	Cost (DKK)
LGB window=90d no-decay	12,996,242

Phase 2: Model Comparison (best recency from Phase 1)

Six model architectures evaluated with train_days=120, decay=12d:

Rank	Model	Cost (DKK)	MAE	Train Time
1	SoftProbV2 (LGB hierarchical)	12,228,236	1.455	156s
2	XGBoost	12,706,983	1.455	69s
3	LightGBM	12,812,760	1.407	46s
4	Blend (LGB+XGB+CB avg)	12,827,883	1.411	131s
5	CatBoost	13,070,234	1.408	49s
6	ElasticNet	13,228,950	1.539	65s

Note: RandomForest, ExtraTrees, and Stacking were removed due to memory constraints on the 16GB machine. RF alone took 100+ minutes without `max_depth` limits due to fully-grown trees on 1.4M training rows.

Phase 3: Safety Buffer Optimization

Buffer sweep on top 3 models. Higher buffers continued to reduce total cost because the stockout penalty (1.5x) dominates:

Baseline best: SoftProbV2 buf=1.40 -> **11,868,433 DKK** (waste: 4,203,070, stockout: 5,110,242)

Shelf-Life Data Augmentation Pipeline

Problem

The production training used a flat `WASTE_FRACTION = 0.3` for all items. As demonstrated in the demo evaluation (April 2), this overstates waste costs for long-shelf-life items.

Challenge: 12,348 Items Need Classification

The demo dataset had ~2,285 items pre-classified. The production dataset has 12,348 unique training items, most with Danish-language titles and no existing shelf-life metadata.

Two-Stage Classification Pipeline

Stage 1: Rule-Based Classifier

A keyword-matching classifier covering 20+ Danish food categories:

Category	Keywords	Shelf Life
Alcoholic beverages	ol, fadol, vin, shots, gin, vodka...	365d
Hot drinks	kaffe, espresso, latte, chai...	365d
Pizza	pizza	2d (fresh) / 90d (frozen if gap>=14d)
Kebab/wraps	kebab, durum, pita, falafel...	1d (fresh) / 60d (frozen)
Fresh salads	salat, grontsag, tomat...	3d
Frozen fried	pommes, fries, nuggets...	90d

Result: **7,127 items matched (57.7%)** with high/medium confidence.

Stage 2: Parallel Haiku Agent Classification

The remaining 5,221 items were split into 10 batches of ~523 items. Each batch was dispatched to a Haiku agent running in parallel, with instructions to classify based on Danish title and ordering cadence (`max_gap_days`).

Classification categories: - **fresh_perishable** (1-2d): fresh sandwiches, sushi, open-face dishes - **short_shelf** (3-7d): bread, pastries, cooked meals, dairy - **medium_shelf** (14-60d): cheese, sauces, frozen foods, processed meats - **long_shelf** (90-365d): beverages, alcohol, canned goods, snacks - **non_food** (9999d): services (haircuts, massages, repairs), gift cards, entrance fees

10 agents completed in ~60 seconds total. All 5,221 items classified.

Post-Processing: Agent outputs used inconsistent `storage_type` values (e.g., “Refrigerated”, “refrigerated/frozen”, “Room Temperature”). A normalization step mapped all variants to 5 canonical types: ambient, refrigerated, dry, frozen, `non_food`.

Final Coverage

Metric	Value
Total training items	12,348
Rule-based (high confidence)	7,127 (57.7%)
Haiku-classified	5,221 (42.3%)
Coverage	100%

Shelf-life distribution:

Bucket	Items
1-2 days (fresh)	1,289
3-7 days (short)	1,948
8-30 days (medium)	2,723
31-90 days (long)	1,474
91-365 days (shelf-stable)	4,334
>365 days (non-food)	580

Shelf-Aware Production Results

Per-Item Waste Fraction

$\text{effective_waste} = 0.3 * \min(1.0, \text{avg_gap_days} / \text{shelf_life_days})$

Non-food items ($\text{shelf_life} \geq 9999$) receive a waste fraction of 0.0.

Phase 1 Recency (Shelf-Aware)

Best config shifted from windowed to full history:

Config	Baseline Cost	Shelf-Aware Cost
window=120d, decay=12d	12,812,760 (best)	10,352,994
full history, decay=6d	12,904,234	10,020,346 (best)

Phase 2 Ranking (Shelf-Aware)

Rank	Model	Cost (DKK)
1	SoftProbV2 (LGB hierarchical)	9,395,177
2	XGBoost	9,986,183
3	LightGBM	10,020,346
4	Blend (LGB+XGB+CB)	10,119,622
5	ElasticNet	10,396,697
6	CatBoost	10,703,792

Overall Best (Shelf-Aware + Buffer Sweep)

LightGBM buf=1.40, full history, decay=6d: **8,415,905 DKK**

Comparative Results

Metric	Baseline (flat waste)	Shelf-Aware	Improvement
Total cost	11,868,433 DKK	8,415,905 DKK	-29.1%
Waste cost	4,203,070 DKK	1,105,724 DKK	-73.7%
Stockout cost	5,110,242 DKK	4,873,454 DKK	-4.6%
Mean cost/store	24,025 DKK	17,036 DKK	-29.1%
Median cost/store	14,192 DKK	9,942 DKK	-30.0%
Best model	SoftProbV2 buf=1.40	LightGBM buf=1.40	—
Best recency	120d window, 12d decay	Full history, 6d decay	—

Reproducibility

All models use `random_state=42` (CatBoost: `random_seed=42`). Results are fully deterministic.

Files

File	Description
<code>notebooks/run_production_training.py</code>	Production training script (shelf-aware)
<code>notebooks/results/experiment_results.json</code>	Experiment results
<code>notebooks/results_baseline_no_shelf_explicit_order_results.json</code>	Baseline experiment results
<code>data/raw/production_shelf_life_train.csv</code>	12,348 items with shelf life classifications
<code>data/raw/production_training_items.csv</code>	Training items with ordering cadence
<code>classify_production_items.py</code>	Rule-based classification script
<code>notebooks/results/phase1_recency.png</code>	Recency comparison chart
<code>notebooks/results/phase2_model_rank.png</code>	Model ranking chart
<code>notebooks/results/phase3_buffer_sweep.png</code>	Buffer sweep chart
<code>notebooks/results/per_store_analysis.png</code>	Per-store distribution

Report generated: April 6, 2026